Continuous Calibration Web App Guidance

Guidance for how to use the Continuous Calibration Web App is given below (under Guidance).

Essential items are marked in red and with an asterisk*. Sublisted items will not be visible unless the parent item is selected. *Italicized* items may be invisible depending on other parameters.

Any time, mole, volume, and path length units can be used. However, units must be consistent between inputs, including upload data. Common examples of units are shown below.

Unit	Example
Time_unit	s, min, h
Moles_unit	mol, mmol, μmol
Volume_unit	mL, L, cm^{-3}, dm^{-3}
Path length_unit	mm, cm

An example of Continuous Calibration Web App use is described below (under Demonstrations).

Guidance

First, one must fill in the parameters as described below.

*Upload

Upload: upload of calibration data.

- ***Select Calibration Data**: uploads calibration data in Excel format (.xlsx). The first row is assumed to be headings, i.e., only data from the second row onward are used for fitting. Number of rows must be equal for all data.
- **Sheet Name**: specifies the sheet name. Default: Sheet1.
- ***Time Column**: specifies the location of the time column. The first column is 1. **Time column unit**: time_unit.
- **Apply Calibration**: specify whether the program should apply the obtained calibration curve to uncalibrated data. Default: off.
  - **Select Experimental Data**: uploads experimental data requiring calibration in Excel format (.xlsx). The first row is assumed to be headings, i.e., only data from the second row onward are used for fitting. Number of rows must be equal for all data.
  - **Sheet Name**: specifies the sheet name. Default is Sheet1.
  - **Time Column**: specifies the location of the time column if applicable. The first column is 1. **Time column unit**: time_unit.

*Parameters

Parameters: required calibration conditions.

- ***Initial Solution Volume**: specifies initial volume of the monitored solution to which analyte solution is being infused. **Unit**: volume_unit.
- **Volume Loss Rate**: specifies loss of monitored solution over time, for example into an instrument. **Unit**: volume_unit time_unit^{-1}. Default: 0 (no loss).
- **+ Species**: adds additional species.
- **– Species**: removes additional species.
- **For each species**:
  - **Species Name**: specifies species name.
  - **Initial Moles**: specifies initial moles of species in monitored solution. **Units: mole_unit**. Default: 0.
  - ***Generation Column**: specifies the location of the intensity column of the continuous calibration data. The first column is 1. **Generation column unit**: intensity_unit.

- ***Application Column***: specifies the location of the intensity column of the uncalibrated experimental data. The first column is 1. **Application column unit**: intensity_unit.
- **Continuous Addition**: specifies continuous addition of a solution.
  - **+ Addition**: adds additional continuous addition.
  - **– Species**: removes additional continuous addition.
  - **For each addition**:
    - **Addition Solution Conc.**: specifies concentration of the solution being continuously added. **Unit**: mole_unit volume_unit^{-1}.
    - **Rate of Addition**: specifies rate at which solution is continuously added. **Units**: volume_unit time_unit^{-1}.
    - **After Time**: specifies time at which continuous addition commences. **Unit**: time_unit. Default: 0
- **Discrete Addition**: specifies discrete addition of a solution.
  - **+ Addition**: adds additional discrete addition.
  - **– Species**: removes additional discrete addition.
  - **For each addition**:
    - **Addition Solution Conc.**: specifies concentration of the solution being discretely added. **Unit**: mole_unit volume_unit^{-1}.
    - **Volume Added**: specifies volume of the discretely added solution. **Unit**: volume_unit.
    - **At Time**: specifies time at which discrete addition occurs. **Units**: time_unit. Default: 0.

Fit Options

Fit Options: means to manipulate the calibration data.

- **Fit Equation**: specifies the type of equation used to fit the calibration curve. Equation options are described below. Default: None.

Name	Equation (+ intercept if selected)
Linear	$a[\text{Species}]$
Logarithmic	$a \times \log_e(b[\text{Species}] + 1)$
Exponential	$a(1 - e^{-b[\text{Species}]})$
Tangential	$a \times \arctan(b[\text{Species}])$
Michaelis-Menten	$\dfrac{a[\text{Species}]}{b + [\text{Species}]}$
Langmuir	$\dfrac{a[\text{Species}]}{1 + b[\text{Species}]}$

  - **Fit Intercept**: fit an intercept. Default: off.
  - **Fit Limit of Fitting**: fit a limit of fitting. Default: off.
    - **Test**: specifies the limit of fitting test. Test options are described below.

Test	Description
RMSE	Root mean squared error (RMSE)
MAE	Mean absolute error (MAE)
R^2	R^2
Runs	Determines if residuals occur in a random sequence
Rainbow	Determines if linear coefficients remain stable across data
Harvey-Collier	Determines if the relationship between predictors and response is linear
Shapiro-Wilk	Determines if residuals follow a normal distribution

    - **Selection**: specifies the criterion for optimizing the test output. Criteria options are described below.

Criterion	Description
min	Minimizes the test output

max	Maximizes the test output
last	Determines the last output which exceeds the selected threshold

- ***Threshold***: specifies the *p* value required for tests to pass. Default: 0.05.
  - **Path length**: specifies the path length of the instrument, for use with spectroscopic absorbance data. **Unit**: path_length_unit.
- **Smoothing Equation**: specifies the type of equation used for smoothing. Equation options are described below. Default: concave GAM.

Equation	Description
Monotonic GAM	Monotonic generalized additive model (GAM)
Concave GAM	Concave generalized additive model (GAM)
Savitsky-Golay	Savitsky-Golay filtering
None	No smoothing equation

- ***Window***: specifies number of data points for Savitsky-Golay smoothing. Default: 11.
- **Limit of Detection Standard Deviations**: specifies the number of standard deviations for calculating the limit of detection.

Data Manipulation

Data manipulation: means to manipulate the calibration data.

- **Breakpoint Time Limit**: specifies time limit in which a delay to continuous addition may occur following the specified “After Time”. **Unit**: time_unit. Default: 0.
- **Diffusion Delay**: specifies time required for complete discrete addition diffusion. **Unit**: time_unit. Default: 0.
- **Zero**: zeros all data to the first fitted concentration datum. Default: off.
- **Smoothing Window**: specifies number of data points for rolling average. Default: 1 (no smoothing).
- **Interpolation Multiplier**: specifies fraction of points to use for fitting calibration curve. Default: 1 (no interpolation).

Output

- Once the above parameters have been filled in, the data can be calibrated and/or downloaded, using “Calibrate” or “Download Calibration” as described below. This changes the tab to the “Output” tab, eventually revealing results.
- **Calibrate**: yields results online, shown as the obtained fitted parameters and figures showing the calibration upon the inputted data. These figures can be downloaded in SVG, PNG, EPS and PDF formats using the drop-down format button and download button found below the figures.
- **Download Calibration**: downloads results as described below.

Tab	Description
InputParams	Inputted parameters
OutputParams	Outputted parameters, including fit equation parameters and quality-of-fit metrics
GenerationData	Calibration curve data
ApplicationData	Calibrated experimental data

- Fitting typically takes ~1–10 s, depending on the amount of data, chosen fitting parameters and local internet speed.

Demonstration

An example fitting is shown for simulated experimental data. These data are available as “CC_example_sim.xlsx”.

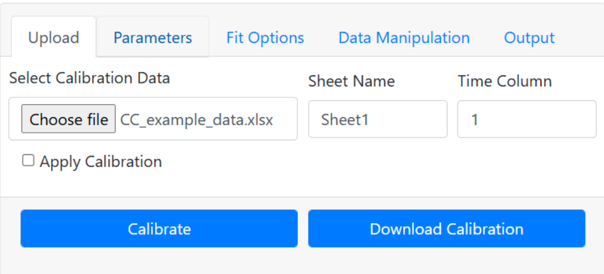
In the simulation, “Species 1” was continuously calibrated by infusing a 1 mol L^{-1} addition solution of “Species 1” at 0.00002 L min^{-1} into a monitored solution of initial volume 0.01 L. Addition occurred from 1 min to 10 min, and 60 spectra were collected each min. The simulated profile used the equation $\mathrm{Intensity} = 100 \times \left(1 - e^{-30[\mathrm{Species\ 1}]}\right) + \mathrm{noise}$, with intensity unit AU.

The resulting data were fit assuming a linear equation with no intercept. The limit of fitting was determined as the concentration at which maximum R^2 was obtained. A concave GAM was also fit.

These data were input into the program as shown below. Units are shown in brackets for clarity but were not put into the program.

Upload

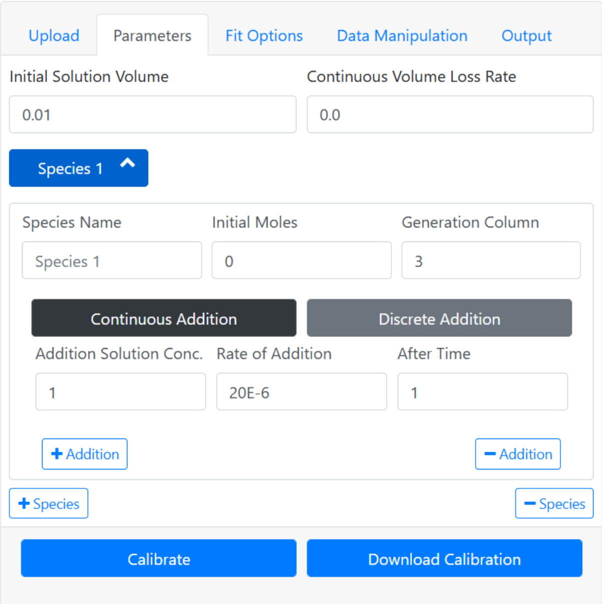
- **Select Calibration Data**: CC_example_sim.xlsx.
- **Sheet Name**: Sheet1 (default).
- **Time Column**: 1 (time column in min).
- **Apply Calibration**: off (default).



Parameters

- **Initial Solution Volume**: 0.01 (L).
- **Volume Loss Rate**: 0 (L min^{-1}, default)
- **Species**:
  - **Species Name**: Species 1 (default).

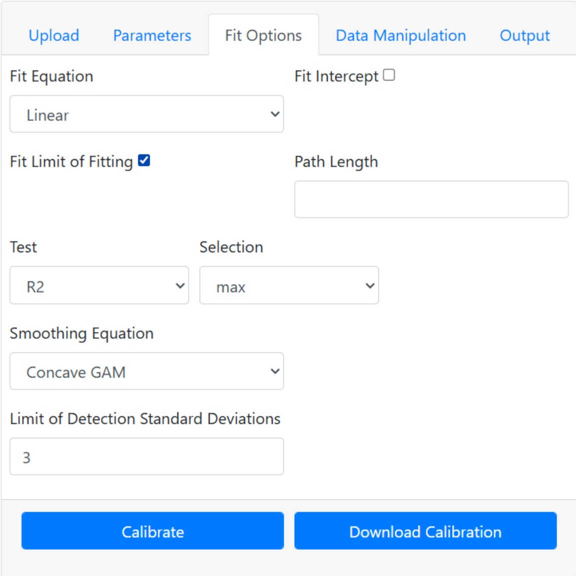
    - **Initial Moles**: 0 (mol, default).
    - **Generation Column**: 3 (generation column in AU).
    - **Continuous Addition**:
        - **+ Addition**.
        - **Addition Solution Conc.**: 1 (mol L^{-1}).
        - **Rate of Addition**: 20E-6 (L min^{-1}).
        - **After Time**: 1 (min).



Fit Options

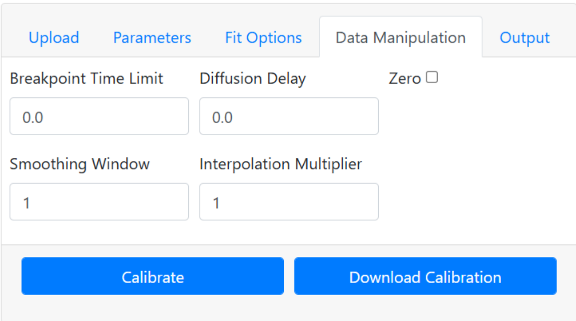
- **Fit Equation**: linear.
- **Fit Intercept**: off (default).
- **Fit Limit of Fitting**: on.
- **Path Length**: left blank (default).
- **Test**: R^2.
- **Selection**: max.

- **Smoothing Equation**: Concave GAM (default).
- **Limit of Detection Standard Deviations**: 3 (default).



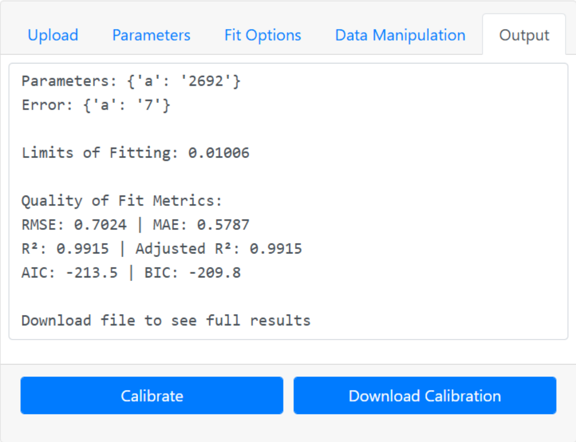
Data Manipulation

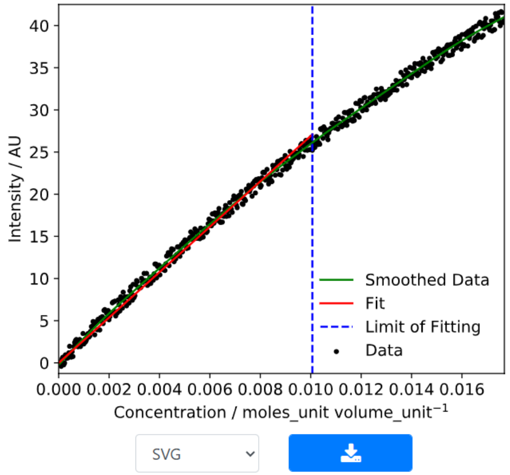
Not used - all left at defaults.



Output

Calibrate:





Download Calibration:

These results are available as “CC_example_fit.xlsx”.

- GenerationData:

	A	B	C	D	E	F
1	Concentration / mole	Intensity / AU	Intensity Standard De	Smoothed Intensity /	Fit Intensity / AU	Fit Residuals / AU
2	0	-0.089498406	0.571426881	-0.248046102	0	-0.089498406
3	3.33322E-05	0.002040318	0	-0.140223727	0.089735389	-0.087695071
4	6.66622E-05	0.125848234	0	-0.032426012	0.179464795	-0.053616562
5	9.99899E-05	-0.419594944	0	0.075335605	0.269188221	-0.688783164
6	0.000133316	0.493024349	0	0.183050457	0.358906307	0.134118042
7	0.000166639	0.670842528	0	0.290705573	0.448617772	0.222224757
8	0.00019996	-0.234619543	0	0.398290297	0.538323256	-0.772942799
9	0.000233279	-0.018337779	0	0.505793209	0.628022762	-0.646360541
10	0.000266596	0.946976662	0	0.61320289	0.71771629	0.229260372
11	0.00029991	0.543204725	0	0.720507929	0.807403839	-0.264199115
12	0.000333222	0.810986519	0	0.827696917	0.897085412	-0.086098893
13	0.000366532	0.583181083	0	0.934758451	0.986761008	-0.403579925
14	0.00039984	1.173164487	0	1.041681897	1.076431269	0.096733217
15	0.000433146	1.965879202	0	1.14845433	1.166094914	0.799784288
16	0.000466449	0.990755916	0	1.255065127	1.255752584	-0.264996668
17	0.00049975	0.686737776	0	1.361502902	1.345404279	-0.658666504
18	0.000533049	1.320926309	0	1.467756275	1.435050002	-0.114123693
19	0.000566346	0.745873809	0	1.573813869	1.524689751	-0.778815942
20	0.00059964	1.943117023	0	1.679664314	1.614323527	0.328793495
21	0.000632933	2.420376062	0	1.785296996	1.703951973	0.716424089
22	0.000666223	2.650325537	0	1.890699044	1.793573807	0.85675173
23	0.00069951	2.770570755	0	1.995859854	1.88318967	0.887381085
24	0.000732796	2.267667055	0	2.100768075	1.972799563	0.294867492
25	0.000766079	1.905543208	0	2.205412356	2.062403486	-0.156860278

- OutputParams:

	A	B	C	D	E	F	G	H	I	J
1	Coefficient(s)	Limit of Fitting	Limit of Detection	RSS	RMSE	MAE	R2	R2_adj	AIC	BIC
2	{'a': 2692.15396327	0.010064346	0.000603525	150.4584	0.702358	0.578711	0.99148	0.991452	-213.521	-209.8

- InputParams:

	A	B	C	D	E	F	G	H	I	J	K	L	M	
1	gen_xl	_sheet_na	gen_t_col	vol0	b_cont_ra	spec_name	mol0	gen_col	dd_sol_co	d_cont_ra	t_cont	d_one_sh	_one_sho	fi
2	<FileStora	Sheet1	0	0.01	0.0	['Species 1	[0.0]	[2]	[1.0]	[[2e-05]]	[[1.0]]	[None]	[None]	Line

The linear equation had a gradient of 2690 AU mol^{-1} L and limit of fitting of 0.0101 mol L^{-1} .